BIRTH DEFECTS

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- A birth defect is an abnormality of structure, function or metabolism (body chemistry) present at birth that results in physical or mental disabilities or death.

Abnormal development . { Miscarriage / Stillbirth

- ***Miscarriage.***
  - (spontaneous abortion) the natural ending of a pregnancy before the embryo or fetus could possibly survive
- ***Stillbirth .***
  - the natural ending of a pregnancy after 20 weeks

CAUSES OF BIRTH DEFECTS

- Both genetic and environmental factors, or a combination of these factors, can cause birth defects.
- However, the causes of about 70 percent of birth defects are unknown.

- **A congenital anomaly** is a structural abnormality of any type that is present at birth.

- Congenital anomalies may be induced by genetic or environmental factors.

- Congenital anomalies may be single or multiple and of minor or major clinical significance.

- During the first 2 weeks of development, teratogenic agents usually kill the embryo or have no effect.

- During the organogenesis period (3rd – 8th weeks), teratogenic agents may cause major congenital anomalies.

- During the fetal period (9th week – 9th month) teratogens may produce morphological and functional abnormalities, particularly of the brain and eyes.

CAUSES OF CONGENITAL ANOMALIES

- **-Genetic factors** such as chromosomal abnormalities and mutant genes.
- **-Environmental factors** e.g.: the mother had German measles in early pregnancy will cause abnormality in the embryo.
- **-Combined** genetic and environmental factors (mutlifactorials factors).

TYPES OF ABNORMALITIES

1-Malformations:

occurs during the formation of the structures of the organ (during organogenesis) results in partial or complete non formation or alterations in the normal structure.

This occurs in the 3rd to the 8th week of gestation.

Example: Cleft lip and or cleft palate.

2-Disruptions:

results in morphological change of the already formed structure due to exposure to destructive process. e.g: vascular accidents leading to intestinal atresia, amniotic band disruption.

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1 cm

3-Deformations:

due to mechanical forces that affect a part of the fetus over a long period.

e.g: talipes equinovarus deformity.

4-Syndrome:

is a group of anomalies occurring together due to a common cause .

e.g.: Down Syndrome

1- GENETIC FACTORS

The genetic factors leading to congenital anomalies may be due to :

- 1- Chromosomal abnormalities,
- 2- Gene mutations or
- 3- Multifactorial.

1) CHROMOSOMAL ABNORMALITIES

- Occur due to:

  1- late maternal age at the time of pregnancy (leads to chromosomal non-disjunction),

  2- radiation (causes chromosome deletions, translocations or breaks),

  3- viruses as German measles,

  4- autoimmune diseases,

  5- and some chemical agents as anti-mitotic drugs.

- **Chromosomal abnormalities are classified into:**
- **A- numerical and B- structural anomalies**.

- **Chromosomal anomalies:**

- usually result from an error that occurred when an ova or sperm cell was developing.

- As a result of this error, a baby can be born with too many or too few chromosomes, or with one or more chromosomes that are broken or rearranged.

A- NUMERICAL CHROMOSOMAL ANOMALIES

- Are divided into:
- 1- Polyploidy
- 2- Anueploidy

1- POLYPLOIDY :

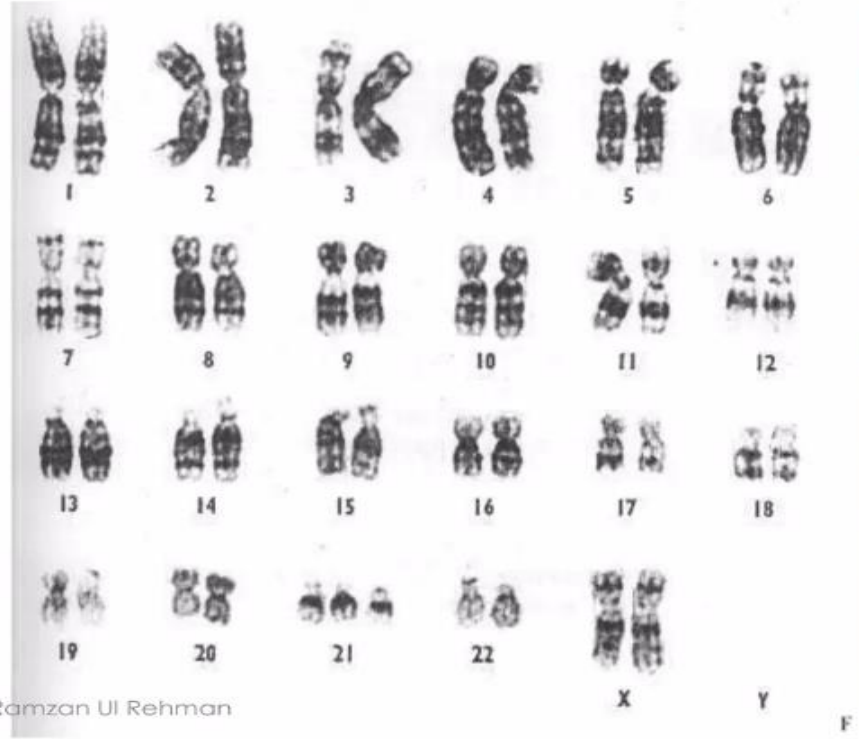
as **triploidy** (a fetus with 69 chromosomes) and **tetraploidy** where the fetus has 92 chromosomes. Polyploidy leads to severe congenital anomalies and early abortion.

2- Aneuploidy :

(one or more chromosomes is added or missed) as in:

Down syndrome (trisomy 21), in which, children have varying degrees of mental retardation, characteristic facial features and, often, heart defects and other problems.

Down syndrome (trisomy 21),



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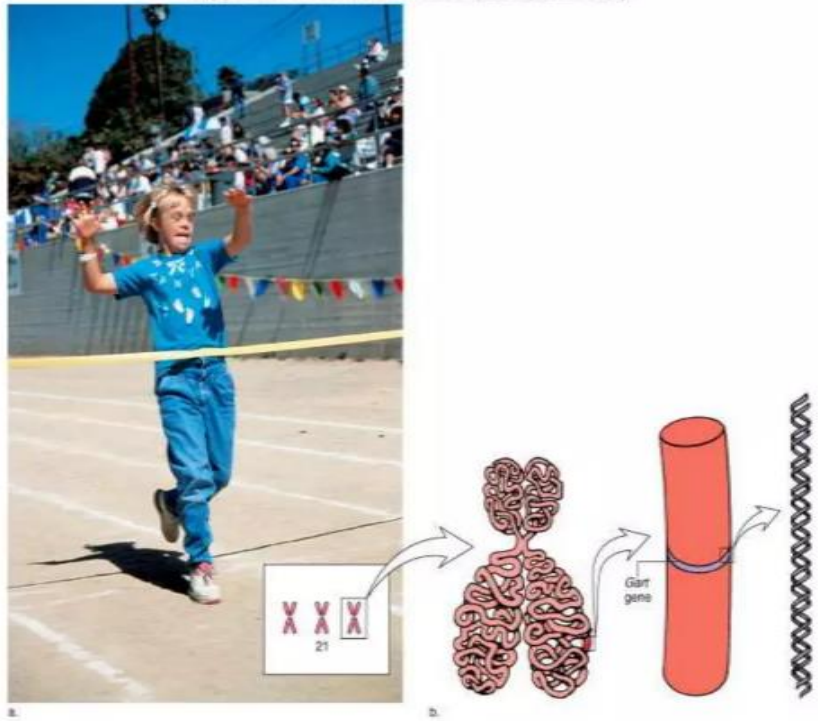

- Edward syndrome (trisomy 18),
- Patau syndrome (trisomy 13),

- **Turner syndrome** (45,X or a female missing one X), and **Klinefelter syndrome** (47,XXY or a male person with an extra X chromosome).

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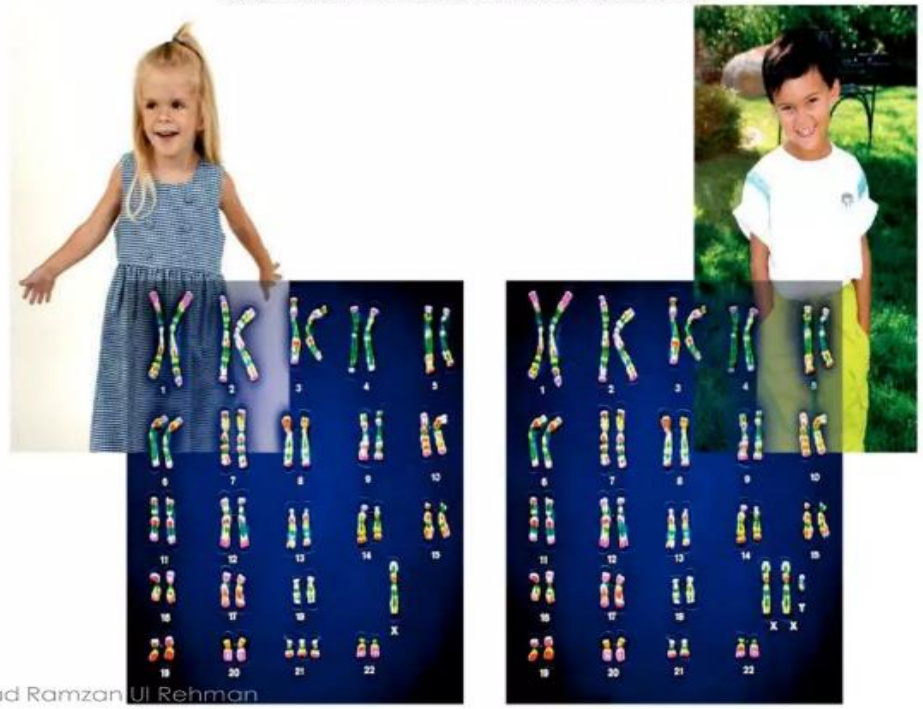


Missing or extra sex chromosomes (X and Y) affect sexual development and may cause infertility, growth abnormalities and behavioral and learning problems. However, most affected individuals

B- STRUCTURAL CHROMOSOMAL ANOMALIES

Include chromosomal :

- 1- deletion,
- 2- duplication,
- 3- translocation,
- 4- inversion,
- 5- ring and
- 6- iso chromosomes.

It may also lead to severe congenital anomalies or fetal death.

2- ENVIRONMENTAL FACTORS:

- Environmental substances that can cause birth defects are called teratogens.
- These include:
  1- Infections,
  2- Radiation,
  3- Chemicals: certain drugs/medications,
  4- Alcohol
  5- Hormones
  6- Maternal disease, Nutritional deficiencies, food poisoning.

ENVIRONMENTAL FACTORS

1) Infectious Agents:

Include a number of viruses:

- Rubella used to be a major problem. It causes cataract, glaucoma, heart defects and deafness.
- Cytomegalovirus :The infection is often fatal and if not meningoencephalitis produce mental retardation.
- Herpes simplex, varicella and HIV.
- Toxoplasmosis
- Syphilis : leads to congenital deafness and mental retardation.

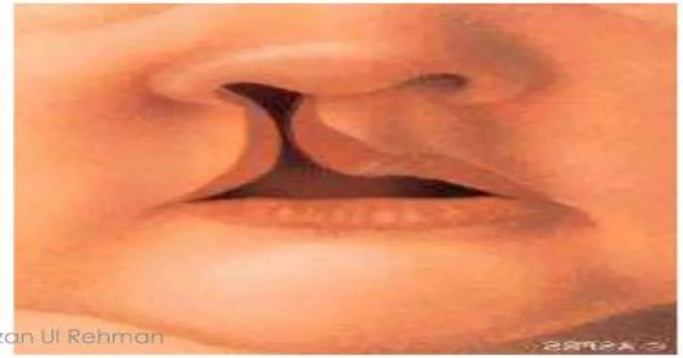
ENVIRONMENTAL FACTORS .

2) Radiation :

Ionizing radiation kills rapidly proliferating cells, producing any type of birth defect depending upon dose and stage of development.

Ex. Atomic bomb on Hiroshima and Nagasaki.

Exposure of the pregnant woman to a large dose of x-ray can lead to microcephaly, spina bifida or cleft palate.



ENVIRONMENTAL FACTORS .

3) Chemical agents:

Many drugs, if given to a pregnant female, can produce congenital anomalies. Ex.:

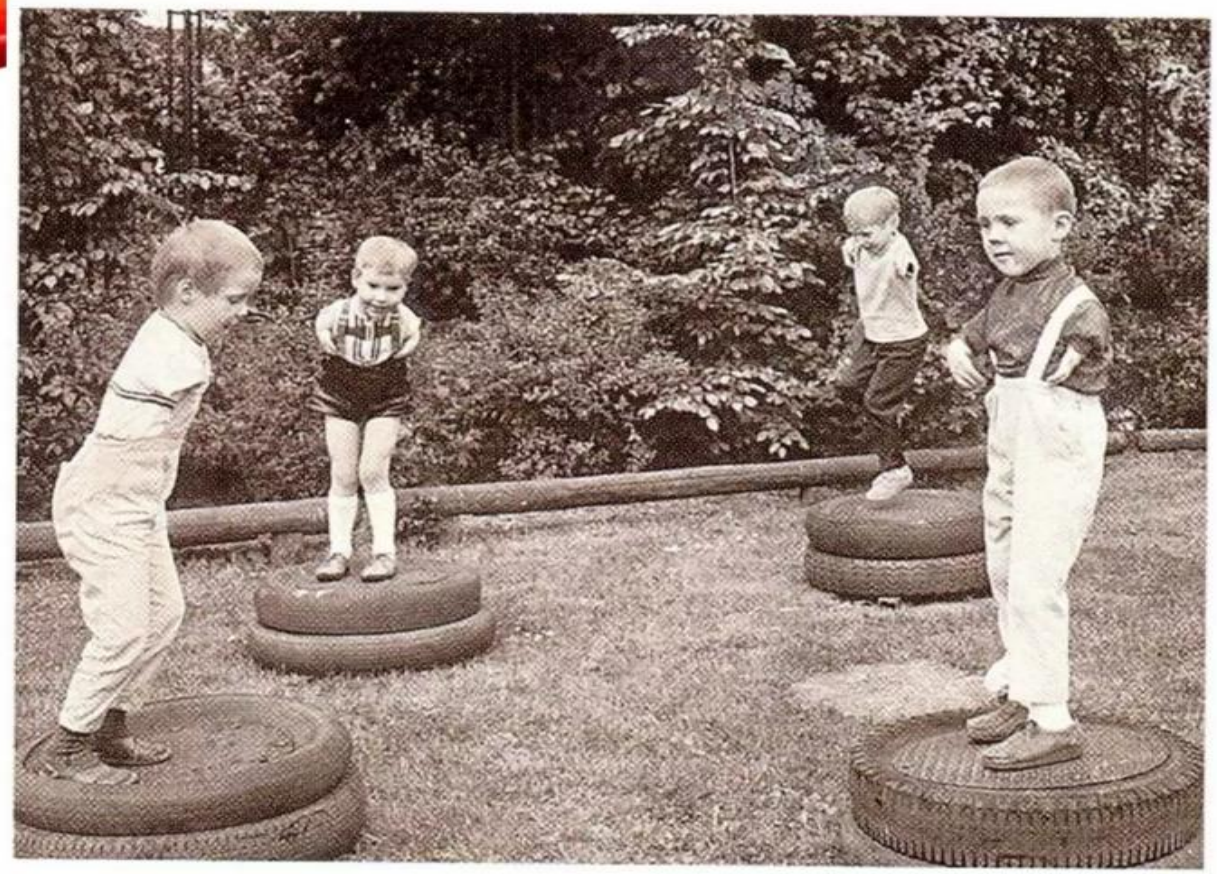
1-Thalidomide (antinauseant sleeping pills) produce limb defects (phocomelia) and heart malformations.

2- Diphenylhydantoin produce facial defects and mental retardation.

3- Tetracycline (bone and teeth anomalies)

4- Aspirin may cause harm in large doses.

5- Cocaine cause birth defect possibly to its effect as a vasoconstrictor that cause hypoxia.



ENVIRONMENTAL FACTORS

4- Alcohol:

Alcohol cause fetal alcohol syndrome.(FAS)

FAS is a pattern of mental and physical birth defects that is common in babies of mothers who drink heavily during pregnancy.

ENVIRONMENTAL FACTORS

5)Hormones:

- Androgenic agents (synthetic progestin to prevent abortion) cause masculinization of the genitalia of female embryos.
- Endocrine hormones as Diethylstilbestrol cause malformation of the uterus, uterine tubes, upper vagina, vaginal cancer and malformed testes OF THE BABY.

ENVIRONMENTAL FACTORS .

6)Maternal Disease:

- Diabetes cause variety of malformations as heart and neural tube defects.

7)Nutritional deficiency:

particularly vitamins deficiency.

8)Heavy metals:

Eg: organic mercury.

3- MULTI-FACTORIAL BIRTH DEFECTS:

- Some birth defects appear to be caused by a combination of one or more genes and environmental exposures. This is called "multi-factorial inheritance."
- In some cases, an individual may inherit one or more genes that make him more likely to have a birth defect if he is exposed to certain environmental substances (such as cigarette smoke).
- These individuals have a genetic predisposition to a birth defect.
- But if the individual is not exposed to the environmental substance before birth, he probably won't have the birth defect.

MULTI-FACTORIAL BIRTH DEFECTS:

Examples of multi-factorial birth defects include:

- 1- Cleft lip / Palate(opening in the lip and/or roof of the mouth)
- 2- Neural tube defects (serious birth defects of the brain and spinal cord, including Spina Bifida and anencephaly)

- Spina bifida (open spine) is one of the most common birth defects.
- Affected babies have varying degrees of paralysis and bladder and bowel problems.
- Both genetic and environmental factors (including insufficient amounts of a vitamin called folic acid) appear to play a role.

3- Heart defects: e.g. Fallot`s tetralogy, Transposition of great arteries

FETAL THERAPY

- The fetus during intrauterine life can receive treatment such as:

1- Fetal transfusion

(administration of blood transfusion to the anemic fetus in thalassemia).

2- Medical treatment

of thyroid dysfunction or congenital adrenal hyperplasia of the fetus.

3- Fetal surgery:

is possible due to advanced ultrasound and surgical procedures eg: repair of hernia of the fetus or in case of hydrocphalus.

4- Stem cell transplantation and gene therapy:

it is possible to transplant stem cells before 18 weeks of gestation of the fetus without rejection because the immunocompetence of the fetus doesn't develop yet